

Player Position Classification

Using Decision Tree and Random Forest

on the FIFA 22 Dataset

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1. Introduction

This project investigates the classification of football players into four positional categories — Goalkeeper (GK), Defender (DEF), Midfielder (MID), and Forward (FWD) — using two machine learning algorithms: the Decision Tree and the Random Forest. The dataset used is the FIFA 22 Complete Player Dataset, sourced from Kaggle, which contains over 19,000 players described by more than 100 attributes.

The primary objective of this study is to compare the performance of both algorithms on the same classification problem, analyze the effect of hyperparameter tuning on each model, and provide a thorough interpretation of the results. The analysis goes beyond simply reporting accuracy scores — it examines per-class performance, overfitting behavior, and the influence of individual hyperparameters on model outcomes.

Player position classification is a naturally challenging problem because certain positions share overlapping skill profiles. For example, attacking midfielders (CAM) exhibit traits similar to forwards, while defensive midfielders (CDM) share characteristics with defenders. This ambiguity makes the problem a rich testbed for comparing model robustness.

2. Dataset Description

The FIFA 22 Complete Player Dataset (players_22.csv) contains 19,239 players and 110 columns. For this project, six numerical features were selected as model inputs, representing the core skill dimensions of each player in the game:

- Pace — overall speed and acceleration rating
- Shooting — finishing and attacking ability rating
- Passing — vision, crossing, and passing accuracy rating
- Dribbling — ball control and agility rating
- Defending — tackling and defensive awareness rating
- Physic — strength, stamina, and physicality rating

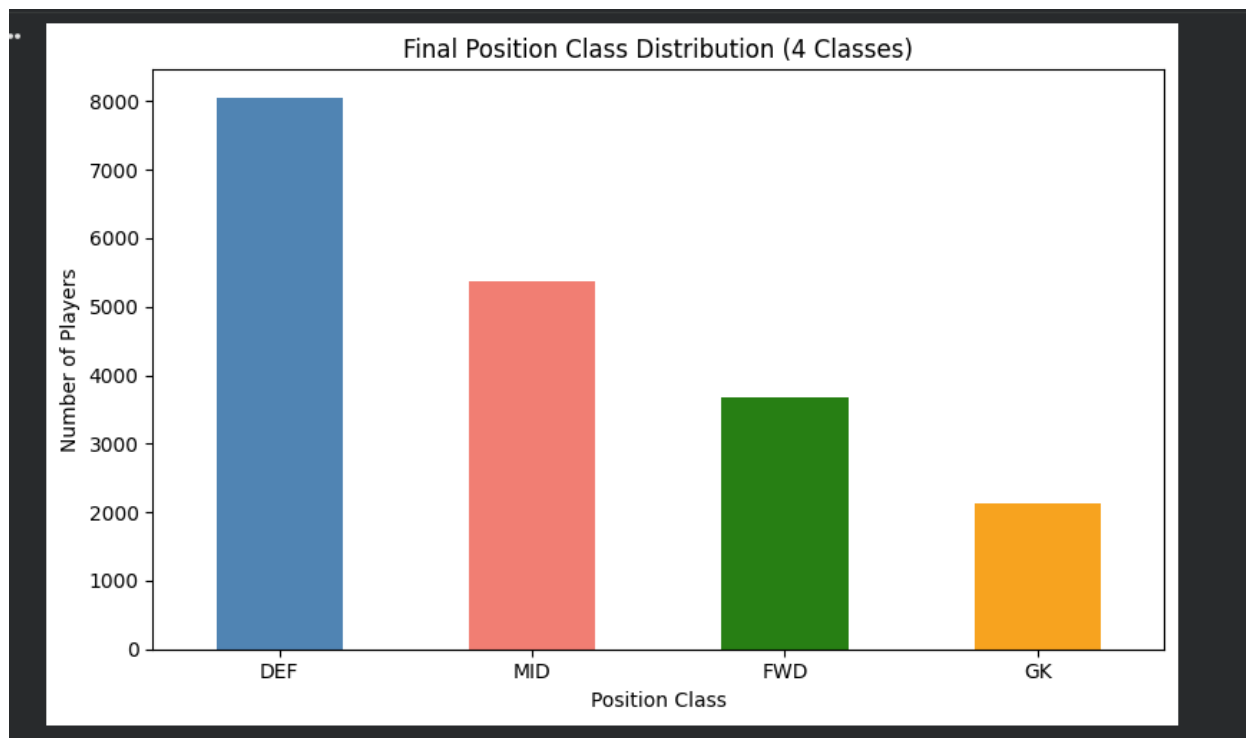
The target variable is the player's primary position, extracted from the player_positions column by taking the first listed position for each player. These raw positions were then mapped into four broad classes: GK, DEF, MID, and FWD. Goalkeepers have missing outfield statistics by design (pace, shooting, etc. are not applicable to them), so these missing values were filled with zero to retain all 2,132 GK records in the dataset.

After cleaning, the final dataset contains 19,239 players distributed as follows:

Position Class	Number of Players	Percentage
DEF	8,059	41.89%
MID	5,368	27.90%
FWD	3,680	19.13%
GK	2,132	11.08%

The dataset is moderately imbalanced, with defenders representing nearly 42% of all players and goalkeepers only 11%. This imbalance is accounted for in the evaluation by reporting per-class F1-scores in addition to overall accuracy.

The data was split into a training set (80%) and a test set (20%) using stratified sampling, ensuring that the class proportions are preserved in both splits. The training set contains 15,391 players and the test set contains 3,848 players.



3. Methodology

3.1 Decision Tree

A Decision Tree is a supervised learning algorithm that recursively partitions the feature space by selecting the split that best separates the classes at each node. The quality of each split is measured using an impurity criterion — either Gini impurity or information entropy. The tree grows until a stopping condition is met, such as reaching a maximum depth or having too few samples to split further.

Decision Trees are highly interpretable and require no feature scaling. However, they are prone to overfitting when allowed to grow without constraints, as they tend to memorize the training data rather than learning general patterns. This makes hyperparameter tuning critical for achieving good generalization.

3.2 Random Forest

Random Forest is an ensemble method that builds a large number of Decision Trees, each trained on a random bootstrap sample of the training data and using a random subset of features at each split. The final prediction is determined by majority vote across all trees. This process introduces two sources of randomness — row sampling (bagging) and feature randomness — which produce diverse trees whose errors partially cancel out when aggregated.

Random Forest inherits the interpretability of individual trees through feature importance scores and is significantly more robust to overfitting than a single Decision Tree. The trade-off is increased computational cost and reduced transparency of the final model.

3.3 Hyperparameter Tuning

Both models were tuned using GridSearchCV with 5-fold cross-validation, which systematically evaluates every combination of specified hyperparameter values across five data splits. The following hyperparameters were explored:

For the Decision Tree:

Hyperparameter	Values Tested	Effect
max_depth	3, 5, 10, 15, 20	Controls tree depth; limits overfitting
min_samples_split	2, 10, 20, 50	Min samples to split a node; higher = simpler tree
min_samples_leaf	1, 5, 10, 20	Min samples at a leaf; higher = smoother predictions
criterion	gini, entropy	Impurity measure for split quality

For the Random Forest:

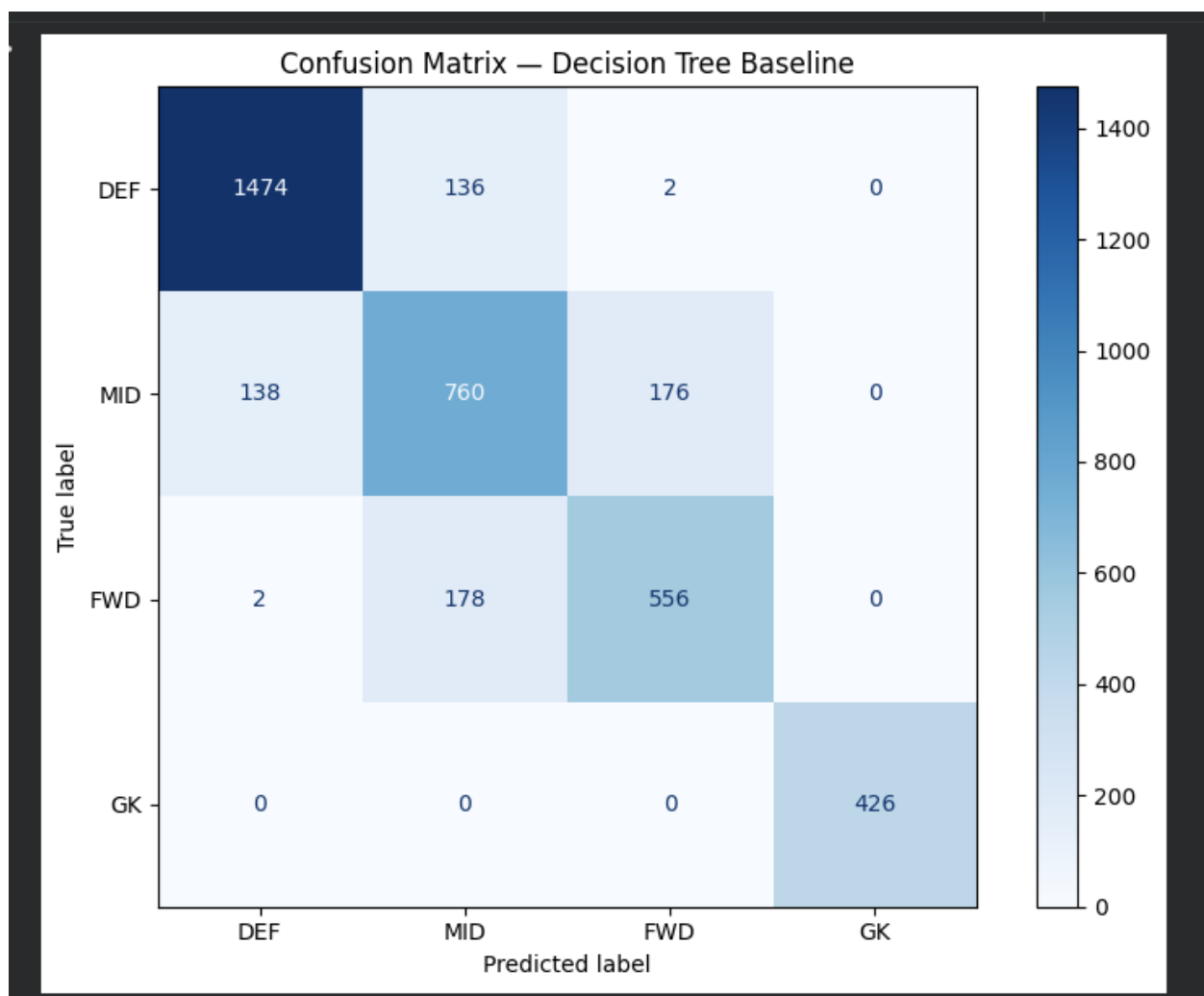
Hyperparameter	Values Tested	Effect
n_estimators	50, 100, 200	Number of trees; more = more stable predictions
max_depth	5, 10, 15, None	Depth of each tree; None = fully grown
max_features	sqrt, log2	Features considered per split; controls tree diversity
min_samples_split	2, 10, 20	Min samples to split a node

4. Results

4.1 Decision Tree — Baseline

The baseline Decision Tree, trained with default hyperparameters, achieved a training accuracy of 99.99% and a test accuracy of 83.58%. The 16.41% gap between training and test accuracy is a clear sign of severe overfitting. The tree grew to a depth of 26 levels with 2,019 leaf nodes — effectively memorizing the training data.

Per-class analysis revealed that GK achieved a perfect F1-score of 1.00, as goalkeepers have a uniquely distinct skill profile (all outfield stats set to zero). DEF achieved an F1-score of 0.91. The hardest classes were MID (F1 = 0.71) and FWD (F1 = 0.76), reflecting the genuine overlap in skill profiles between attacking midfielders and forwards, and between defensive midfielders and defenders.

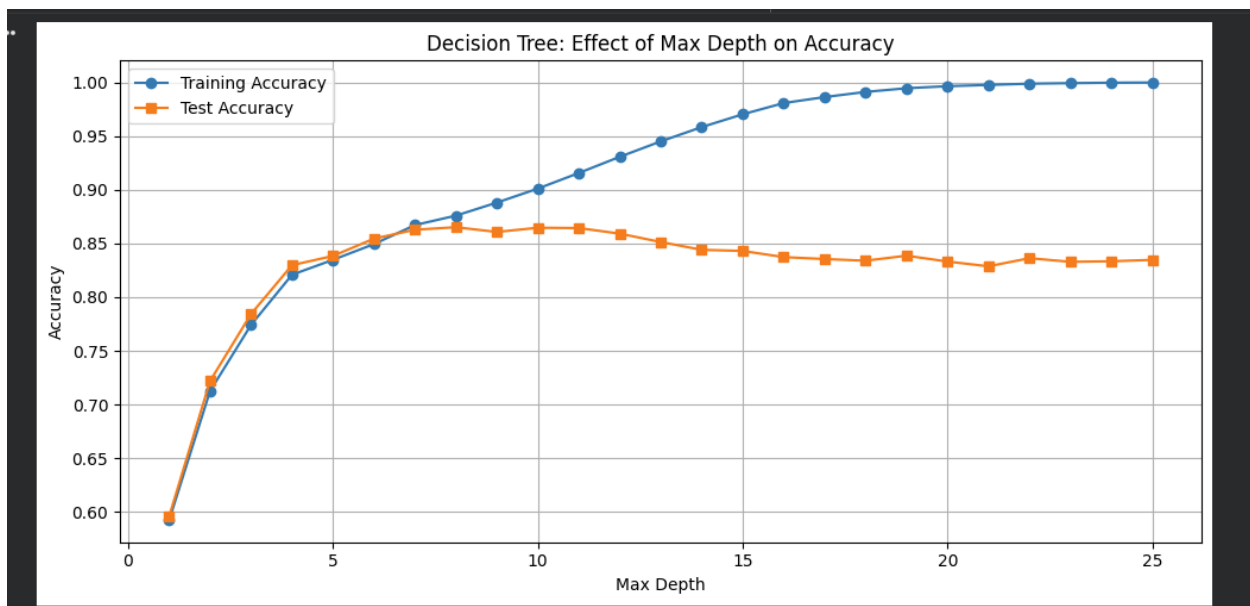


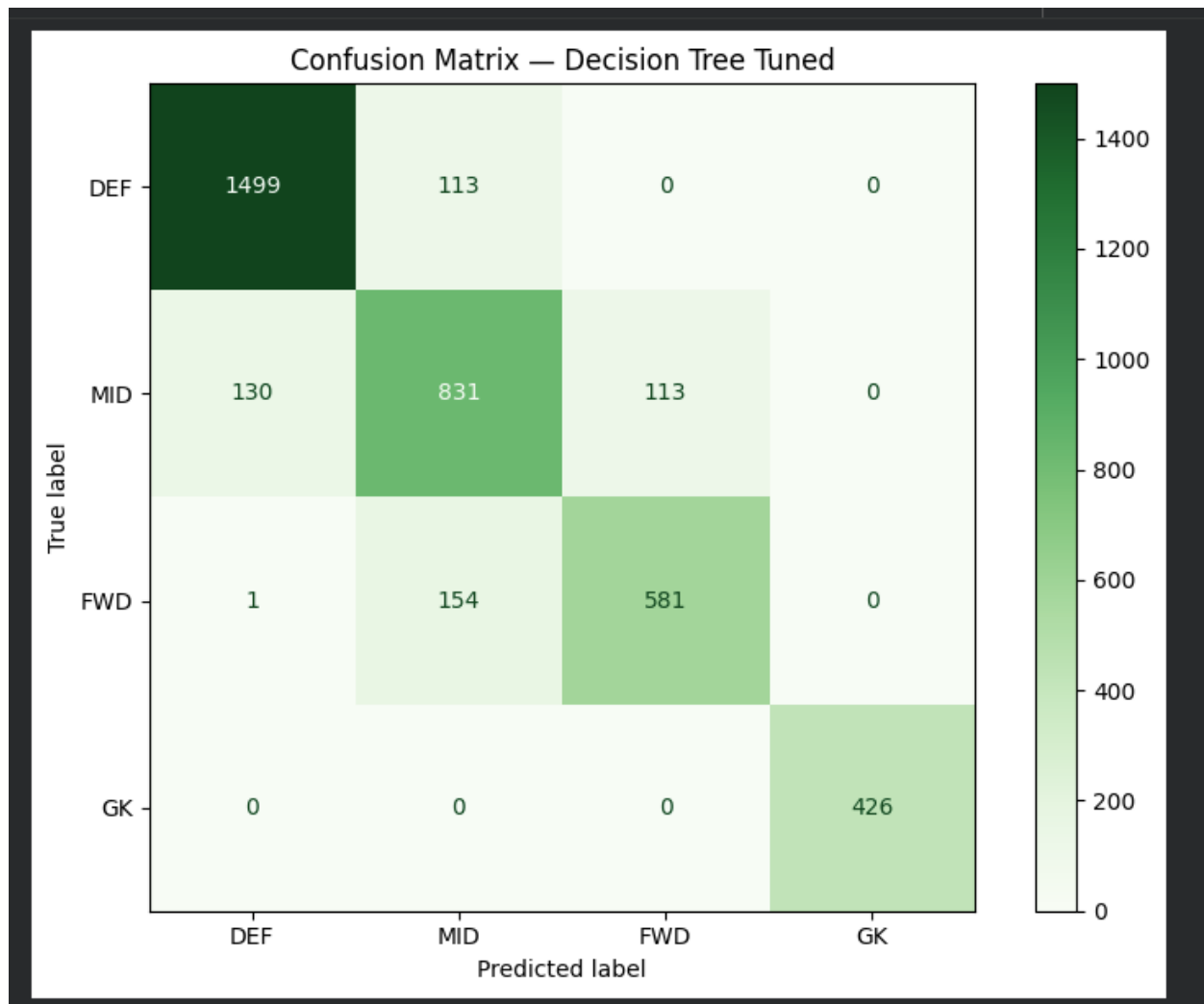
4.2 Decision Tree — Tuned

GridSearchCV identified the following optimal hyperparameters for the Decision Tree: criterion = entropy, max_depth = 10, min_samples_leaf = 5, and min_samples_split = 50. These constraints forced the tree to generalize rather than memorize.

The tuned Decision Tree achieved a test accuracy of 86.72%, an improvement of 3.14 percentage points over the baseline. More importantly, the train/test gap collapsed from 16.41% to just 1.37%, confirming that overfitting was successfully reduced. Per-class F1-scores improved across the board: MID improved from 0.71 to 0.77 and FWD improved from 0.76 to 0.81.

The max_depth exploration plot showed that test accuracy peaked around depth 10 and then plateaued or slightly declined as depth increased further, while training accuracy continued to rise toward 100%. This visually confirms that depth 10 is the optimal sweet spot between underfitting and overfitting.



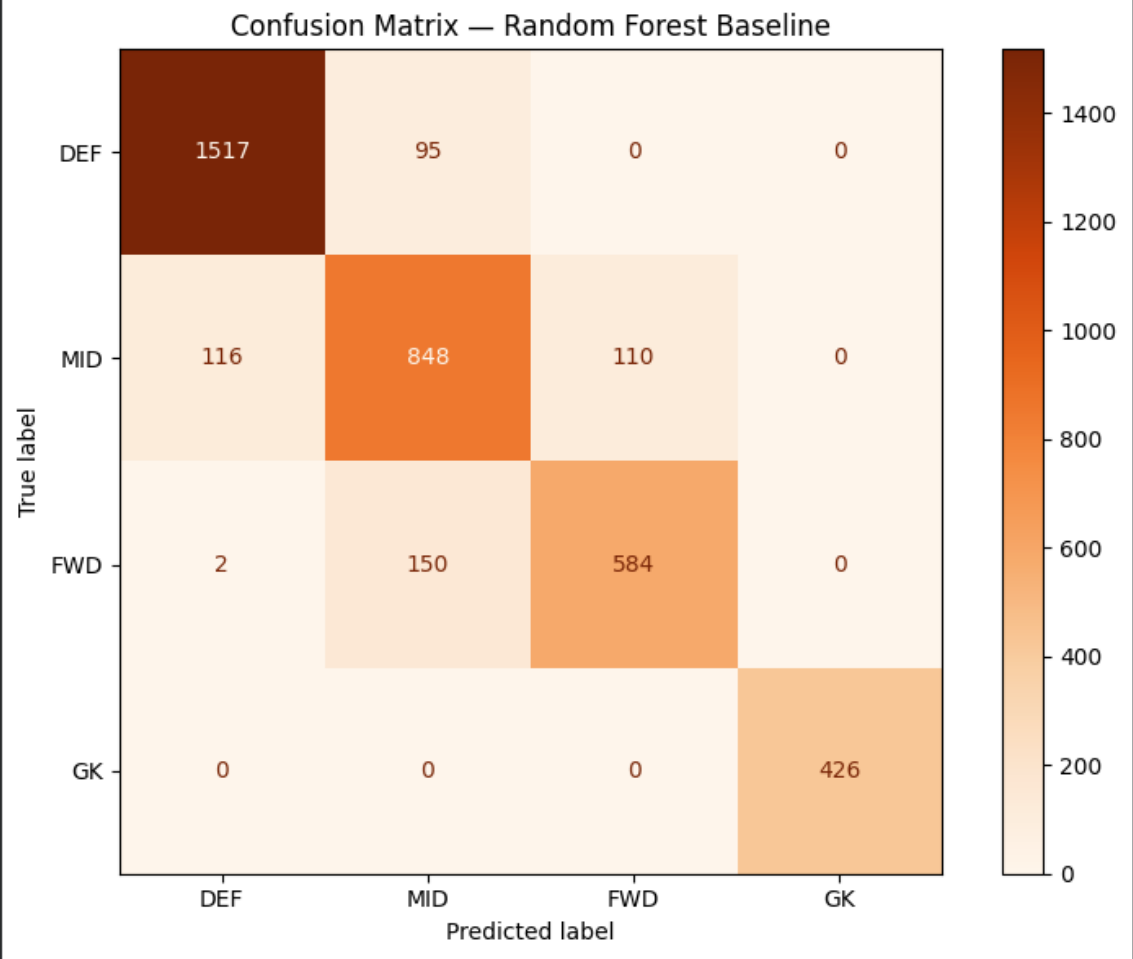


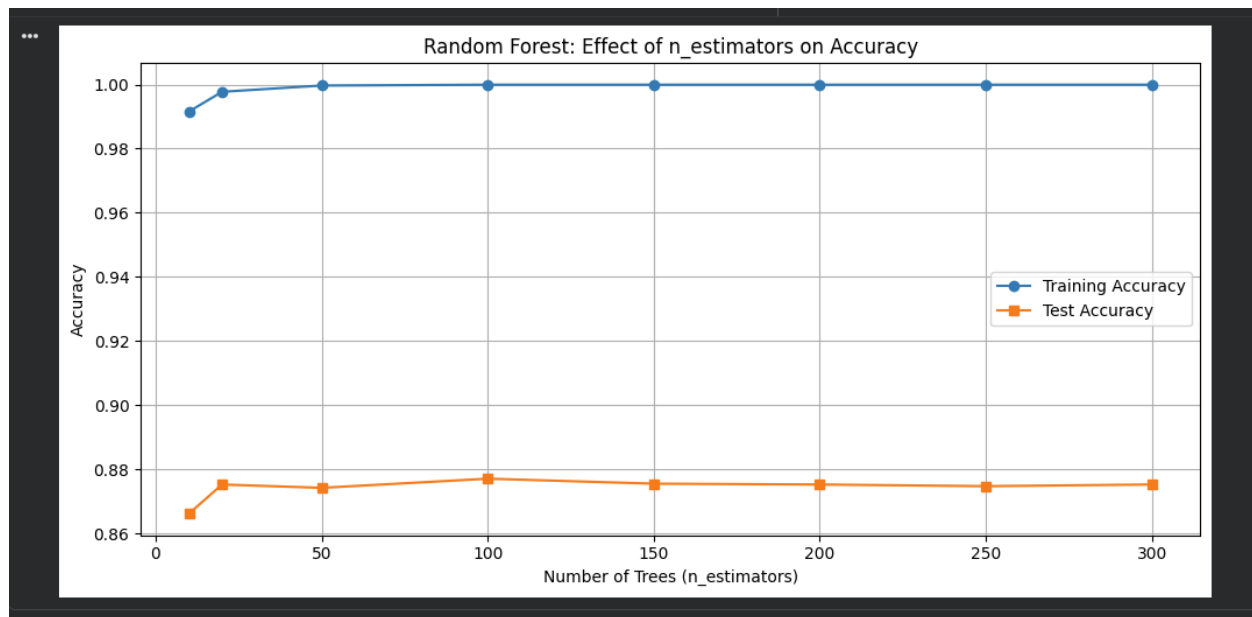
4.3 Random Forest — Baseline

The baseline Random Forest, using default settings (100 trees), achieved a training accuracy of 99.99% and a test accuracy of 87.71%. Notably, the baseline Random Forest already outperformed the tuned Decision Tree (86.72%) without any tuning. This demonstrates the fundamental advantage of ensemble averaging: even without carefully tuned hyperparameters, combining 100 diverse trees produces a more robust classifier than a single optimized tree.

Per-class F1-scores for the baseline Random Forest were DEF = 0.93, FWD = 0.82, GK = 1.00, and MID = 0.78 — all higher than the baseline Decision Tree scores.

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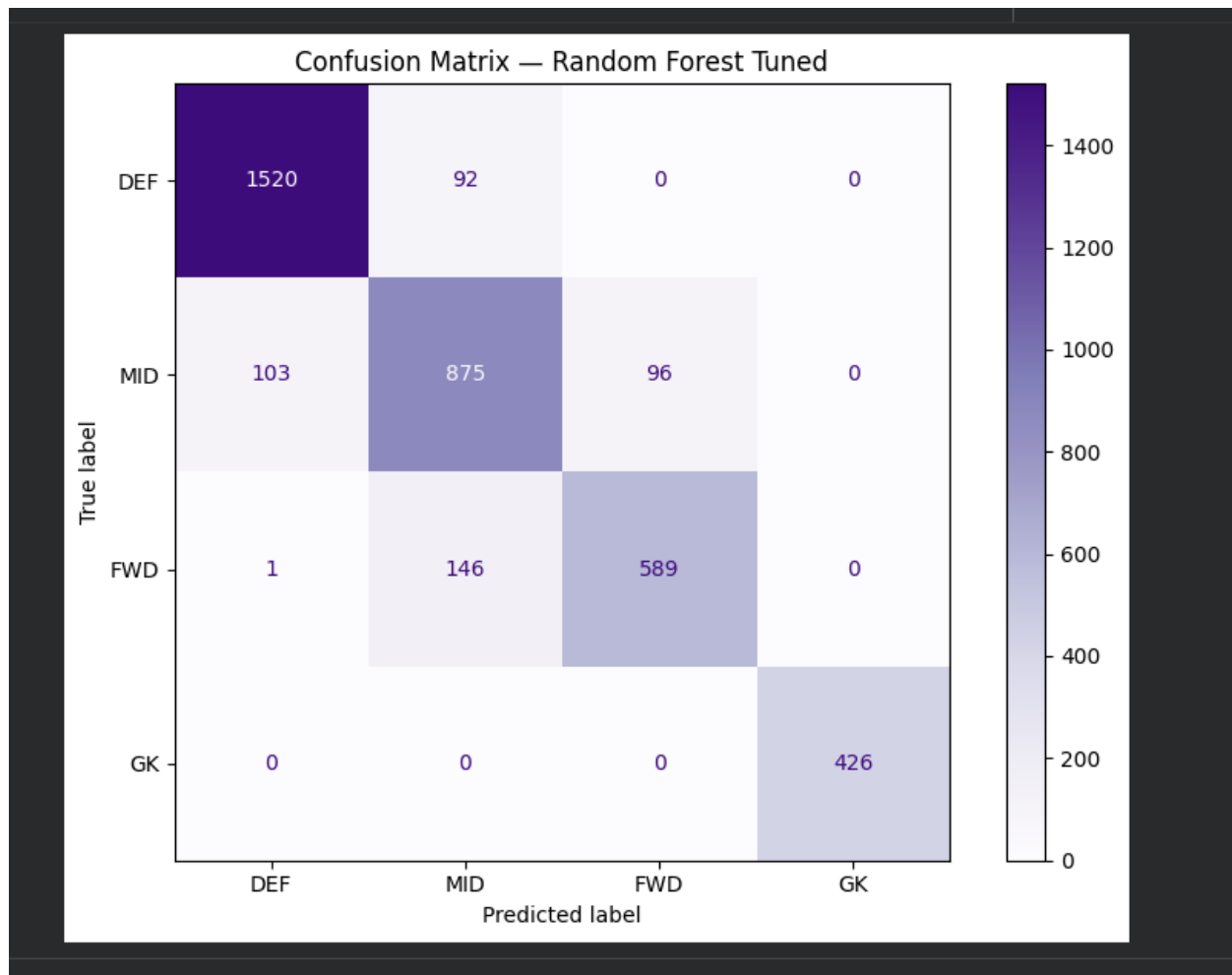


4.4 Random Forest — Tuned

GridSearchCV identified the following optimal hyperparameters for the Random Forest: `n_estimators = 200`, `max_depth = None` (fully grown trees), `max_features = sqrt`, and `min_samples_split = 20`. The tuned Random Forest achieved the best test accuracy of all four models at 88.62%, with a train/test gap of 7.55%.

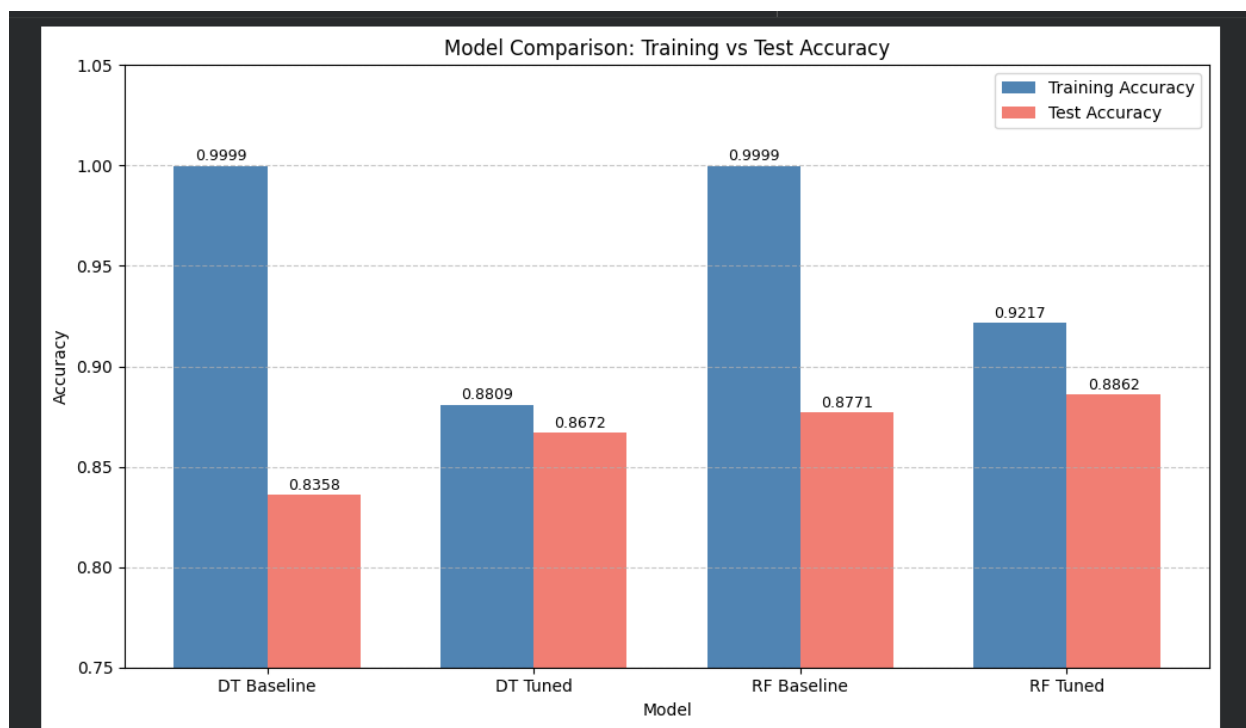
It is worth noting that the best Random Forest used fully grown trees (`max_depth = None`). This is possible because the ensemble averaging effect suppresses the overfitting that would occur in a single unconstrained tree — the variance reduction from combining 200 trees compensates for the high variance of each individual tree.

The `n_estimators` exploration showed that test accuracy stabilized after approximately 50 to 100 trees, with minimal gains beyond that threshold. The improvement from 100 to 200 trees was marginal, confirming the law of diminishing returns in ensemble size.



4.5 Summary of Results

Model	Train Acc	Test Acc	Gap	MID F1	FWD F1
DT Baseline	99.99%	83.58%	16.41%	0.71	0.76
DT Tuned	88.09%	86.72%	1.37%	0.77	0.81
RF Baseline	99.99%	87.71%	12.28%	0.78	0.82
RF Tuned	92.17%	88.62%	7.55%	0.80	0.83

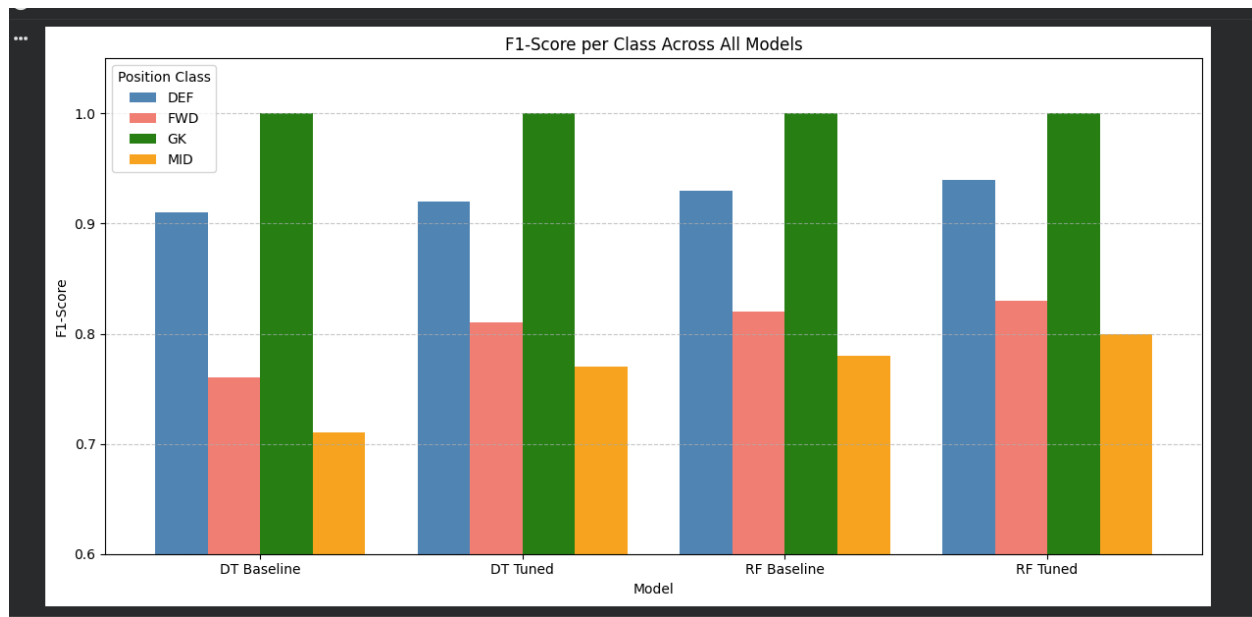


5. Discussion

5.1 Why Random Forest Outperforms Decision Tree

The results consistently show that Random Forest outperforms the Decision Tree at every stage of the comparison — both at baseline and after tuning. The core reason is variance reduction through ensemble averaging. A single Decision Tree is highly sensitive to small changes in the training data, which leads to high variance and overfitting. Random Forest mitigates this by averaging across 200 diverse trees, each trained on a different bootstrap sample and using a different random subset of features at each split.

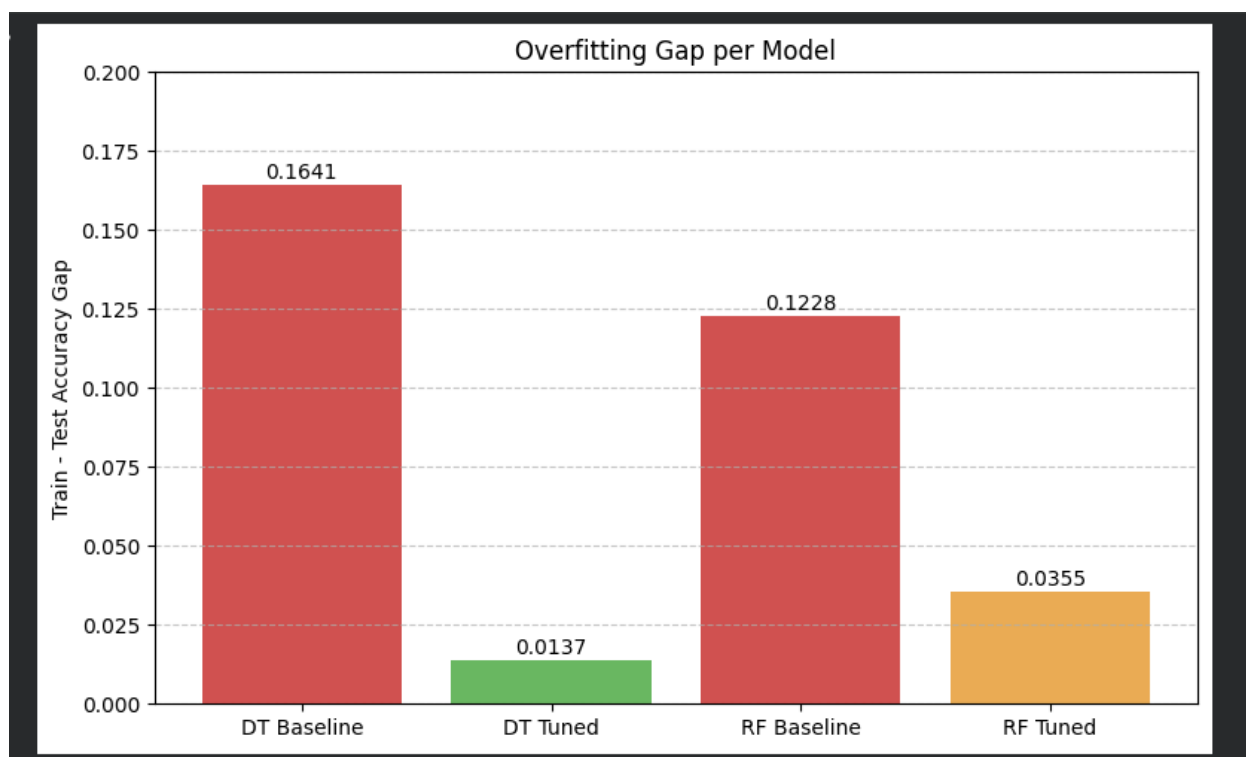
This diversity is key: because no two trees in the forest see exactly the same data or the same features, their individual errors are largely uncorrelated. When their predictions are aggregated, random errors cancel out and the systematic patterns dominate. This is why even the Random Forest baseline outperformed the carefully tuned Decision Tree.



5.2 The Overfitting Story

Both baseline models showed near-perfect training accuracy (99.99%) with significantly lower test accuracy, confirming overfitting. The Decision Tree baseline had the worst gap at 16.41%. Tuning the Decision Tree with `max_depth = 10`, `min_samples_split = 50`, and `min_samples_leaf = 5` dramatically reduced this gap to just 1.37%, demonstrating that depth constraints are the most effective remedy for Decision Tree overfitting.

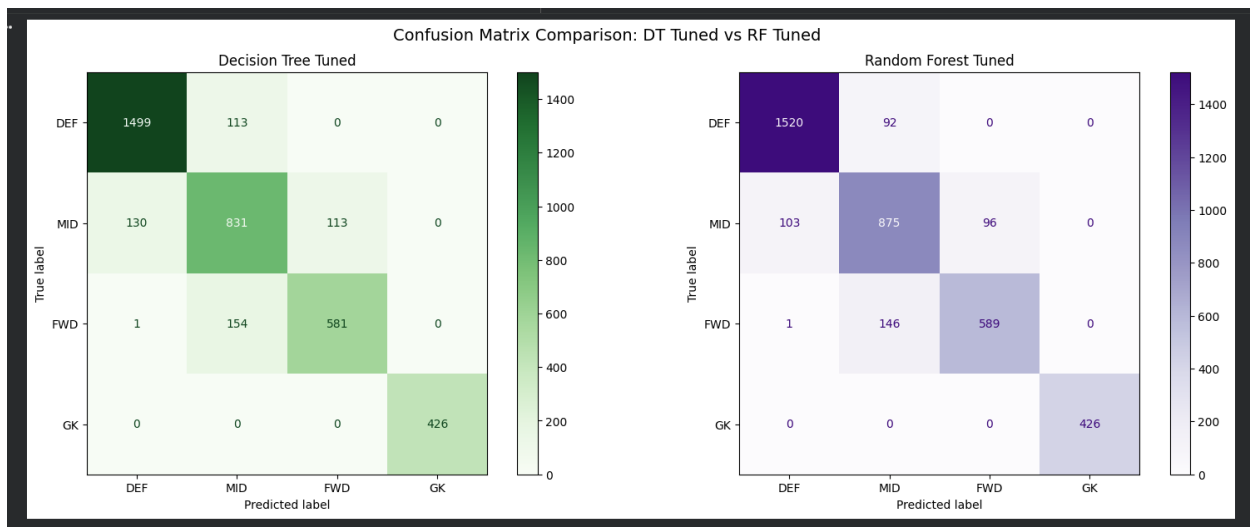
For the Random Forest, the baseline gap of 12.28% was already smaller than the Decision Tree baseline despite using fully grown trees. After tuning, the gap reduced to 7.55%. The Random Forest tolerates unconstrained tree depth better because the ensemble mechanism itself acts as a regularizer. The remaining 7.55% gap suggests that some overfitting persists, but the absolute test performance is still the highest of all four models.



5.3 The MID/FWD Classification Challenge

Across all four models, MID consistently had the lowest F1-score and was the most confused with FWD. This is not a failure of the algorithms — it reflects a genuine ambiguity in the data. Players like attacking midfielders (CAM) and wingers are rated similarly to forwards in terms of pace, shooting, and dribbling. Similarly, defensive midfielders (CDM) share high defending and physic scores with defenders.

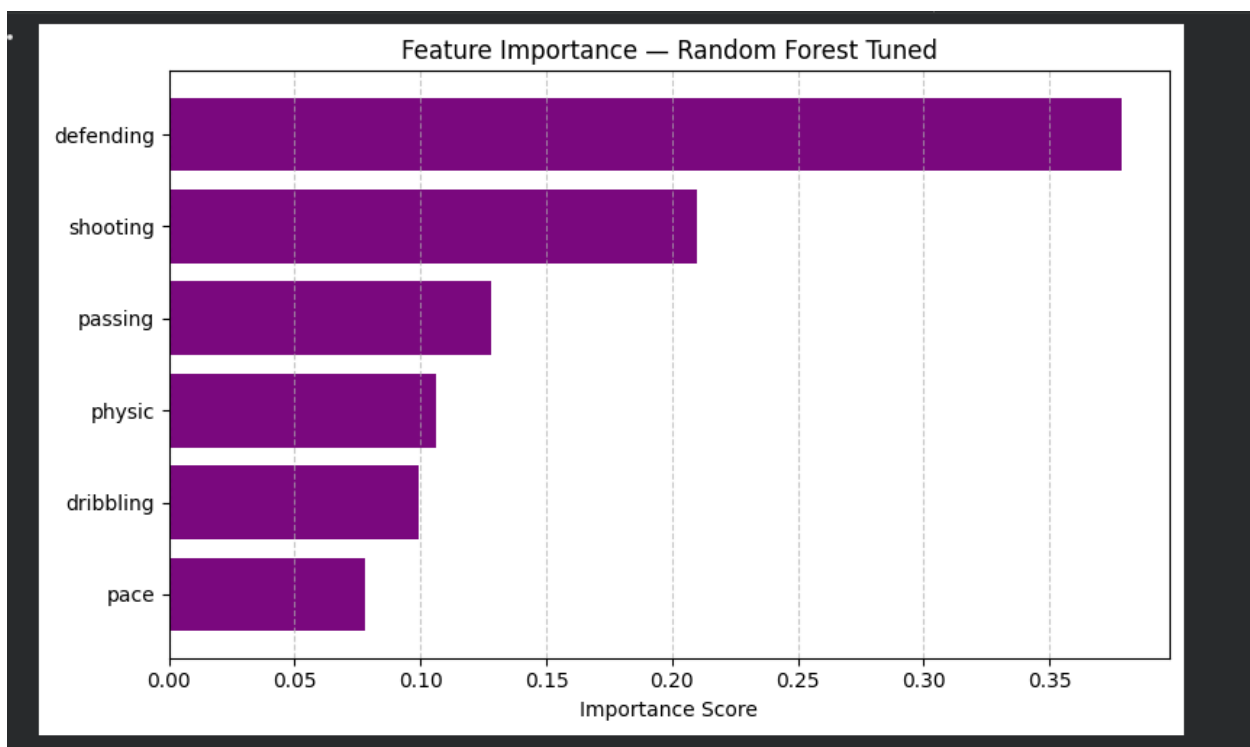
The Random Forest handled this ambiguity better than the Decision Tree in all configurations. With 200 trees each capturing slightly different aspects of the feature space, the forest was better at identifying the subtle statistical patterns that distinguish a midfielder from a forward, pushing MID F1 from 0.71 (DT baseline) to 0.80 (RF tuned).



5.4 Feature Importance Analysis

The Random Forest feature importance analysis revealed that defending was by far the most important feature (importance score approximately 0.38), followed by shooting (approximately 0.21) and passing (approximately 0.13). Pace, dribbling, and physic were the least important features.

This result is highly intuitive from a football perspective. Defending is the single strongest discriminator because only DEF and GK players have high defending scores, while FWD players typically have very low defending ratings. Shooting is the second most important because FWD players have high shooting while GK players have near-zero scores. Passing helps differentiate MID players who tend to have the highest passing ratings across all positions.



6. Conclusion

This project compared Decision Tree and Random Forest classifiers on the task of predicting player positions in the FIFA 22 dataset. Both models were evaluated at baseline and after systematic hyperparameter tuning using GridSearchCV with 5-fold cross-validation.

The key findings are as follows. First, the tuned Random Forest achieved the best overall test accuracy of 88.62%, outperforming the tuned Decision Tree (86.72%) by 1.90 percentage points. Second, the Random Forest baseline (87.71%) already surpassed the tuned Decision Tree, demonstrating that ensemble averaging provides an inherent robustness advantage that tuning alone cannot replicate in a single tree. Third, hyperparameter tuning significantly improved both models — the Decision Tree's overfitting gap was reduced from 16.41% to just 1.37% after constraining max_depth to 10. Fourth, the MID class was the hardest to classify across all models due to overlapping skill profiles with both DEF and FWD players, and the Random Forest handled this ambiguity more effectively. Fifth, feature importance analysis confirmed that defending and shooting are the most discriminative features for position classification, consistent with football domain knowledge.

In conclusion, Random Forest is the recommended algorithm for this classification task due to its superior generalization, robustness to overfitting, and consistent per-class performance. The Decision Tree remains valuable for its interpretability and speed, and its performance can be substantially improved through careful hyperparameter tuning. Future work could explore adding sub-attribute features such as finishing, standing_tackle, and short_passing to further improve classification of the MID and FWD boundary cases.

References

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